

	
4(a)	The CG of a body is the single point at which the entire weight of the body can be considered to act.
4(b)	Let x be the distance of the cg from end A. Fig 4.1, taking moment about the hinge at the wall as axis gives $Wx = 12(3) = 36$ (1) Fig 4.2, taking moment about the hinge at the wall as axis gives $W(3 - x) = 16(3) = 48$ (2)

Qn	Solution
	Equation (2) / (1) $\frac{3 - x}{x} = \frac{16}{12}$ <i>solving</i> $x = 1.29 \text{ m} = 1.3 \text{ m}$ <i>Substituting</i> $x = 1.29 \text{ m}$ into (1) gives $W = 28 \text{ N}$
4(c)	F remains at 16 N. Equation (2) in part 4(b) above is modified by multiplying a $\cos(30)$ term on both sides of the equation, due to the change in perpendicular distance from the pivot. The common factor of $\cos(30)$ cancels out to give back the same equation as the case for 4(b), so $F = 16 \text{ N}$ as x and W remain the same.
5(a)	Pressure at the bottom of the cylinder is given by